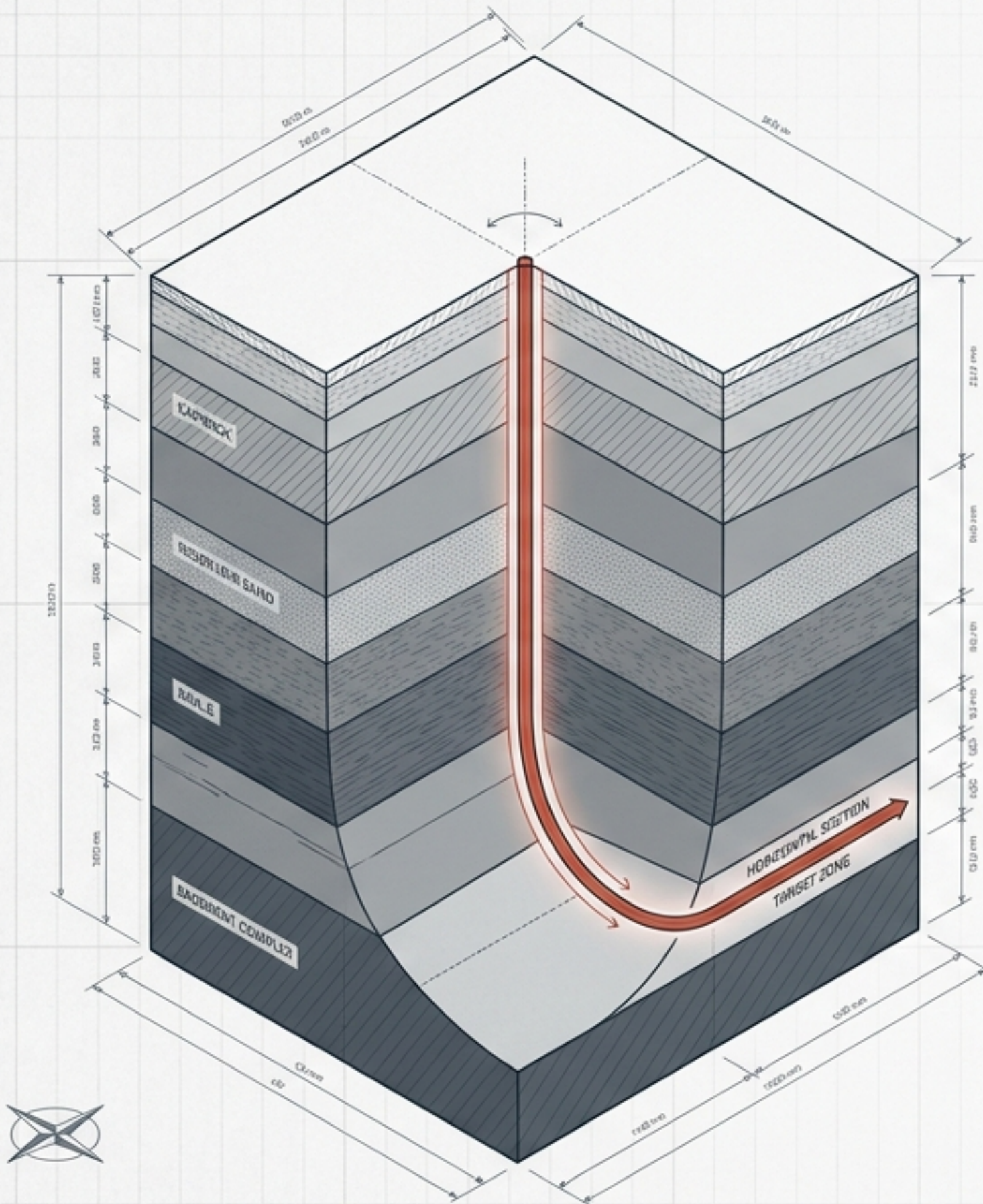


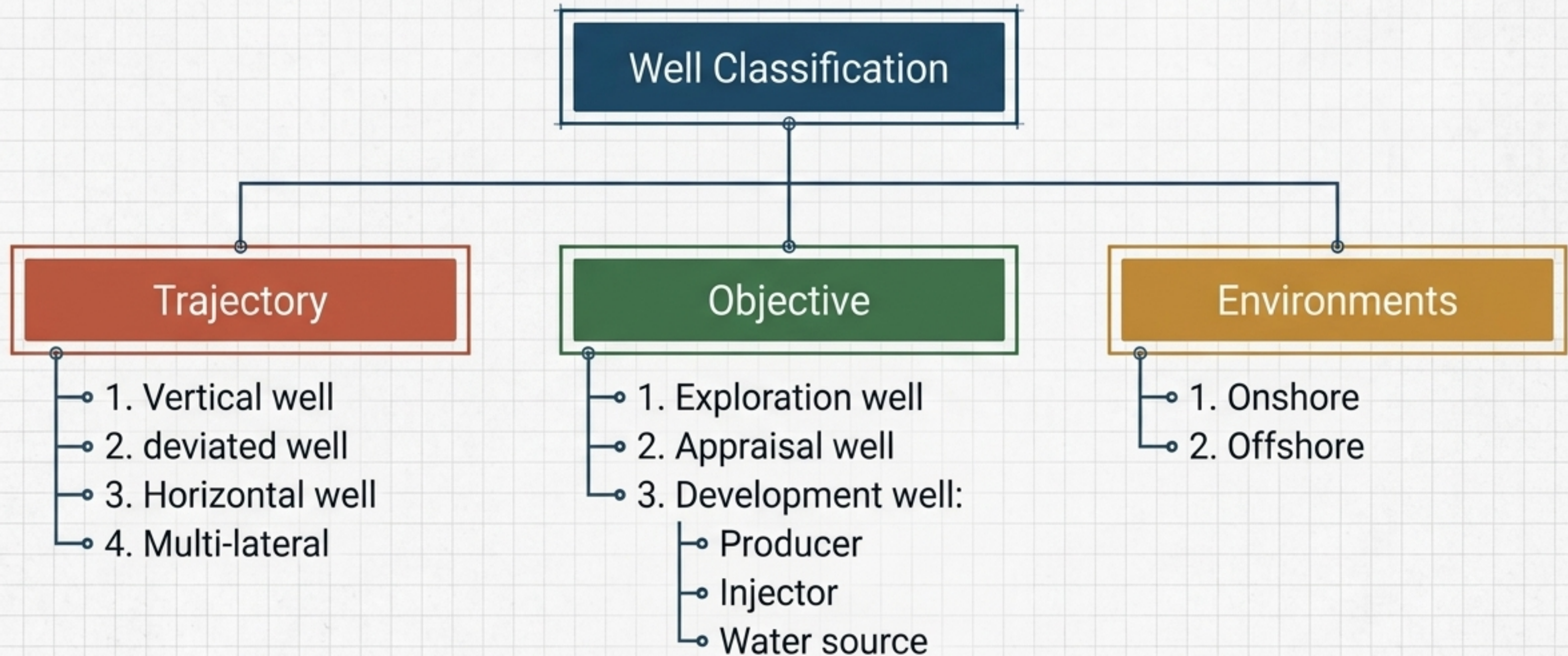
The Architecture of the Subsurface

A Dimensional Taxonomy of Petroleum Well Classification

Engineered and Presented by:
Engineer Sajjad K.Hano



Three Dimensions Define Every Petroleum Well



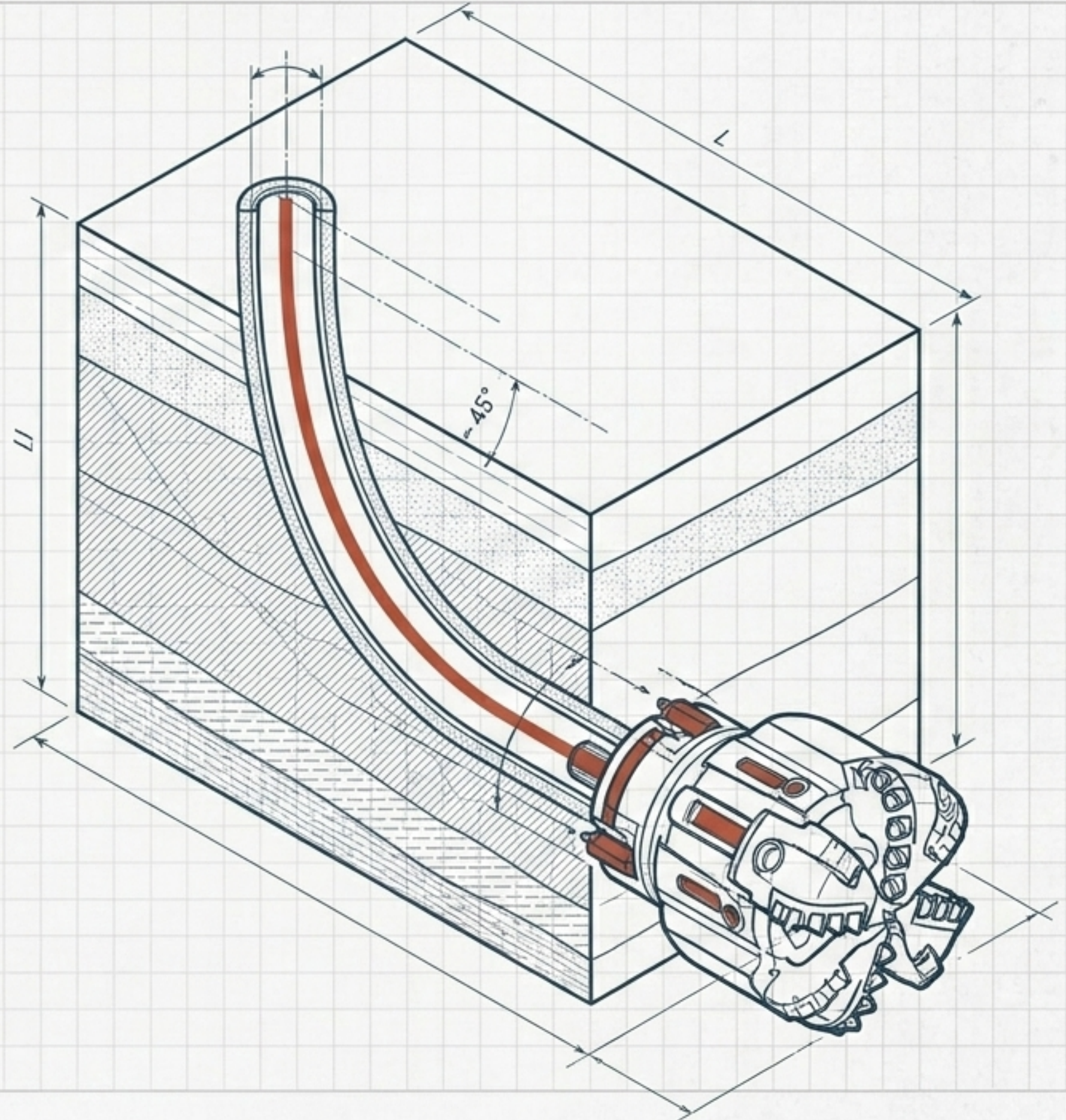
Framework synthesized by Engineer Sajjad K.Hano.

Dimension One: Trajectory

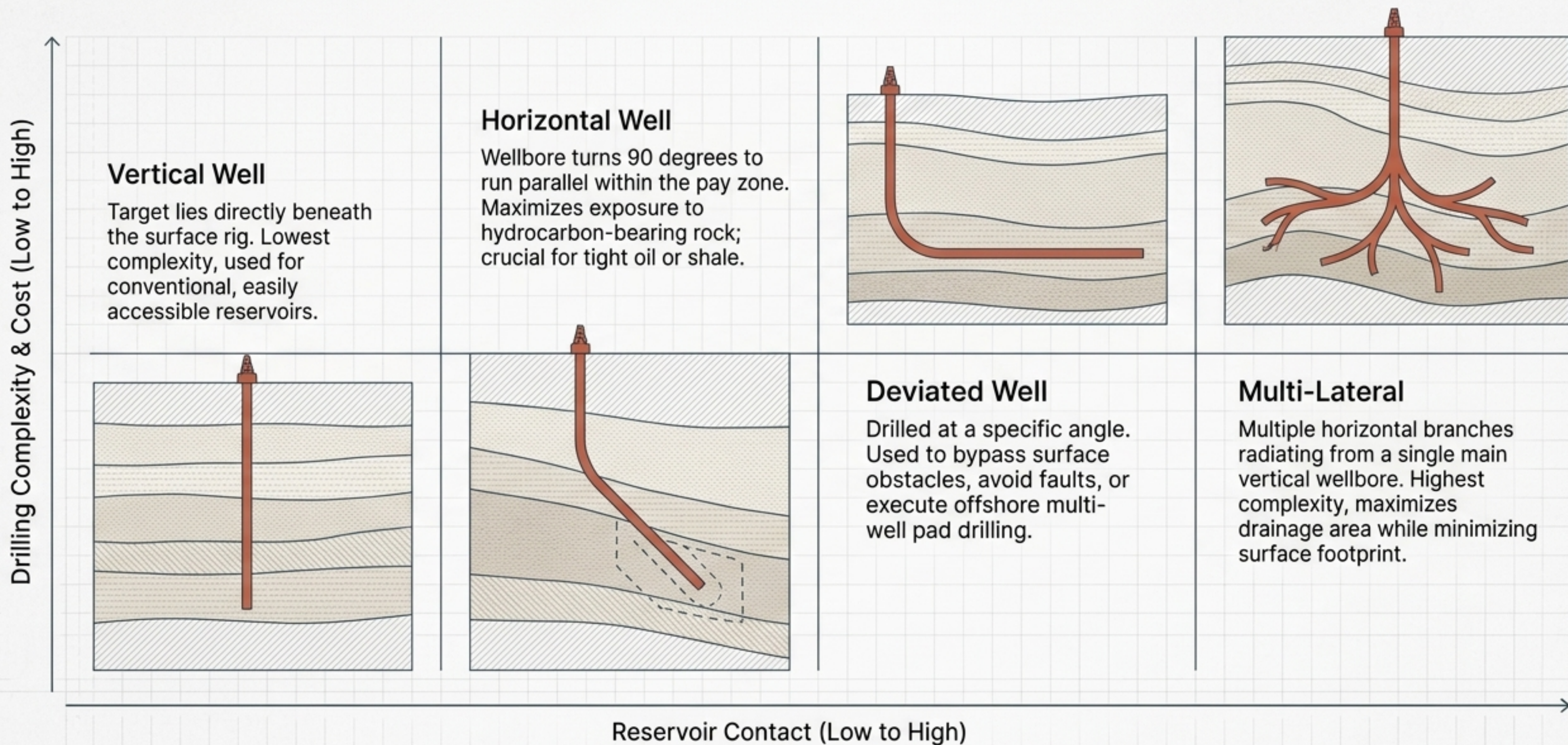
The Geometry of Subsurface Navigation.

Trajectory

- 1. Vertical well
- 2. Deviated well
- 3. Horizontal well
- 4. Multi-lateral

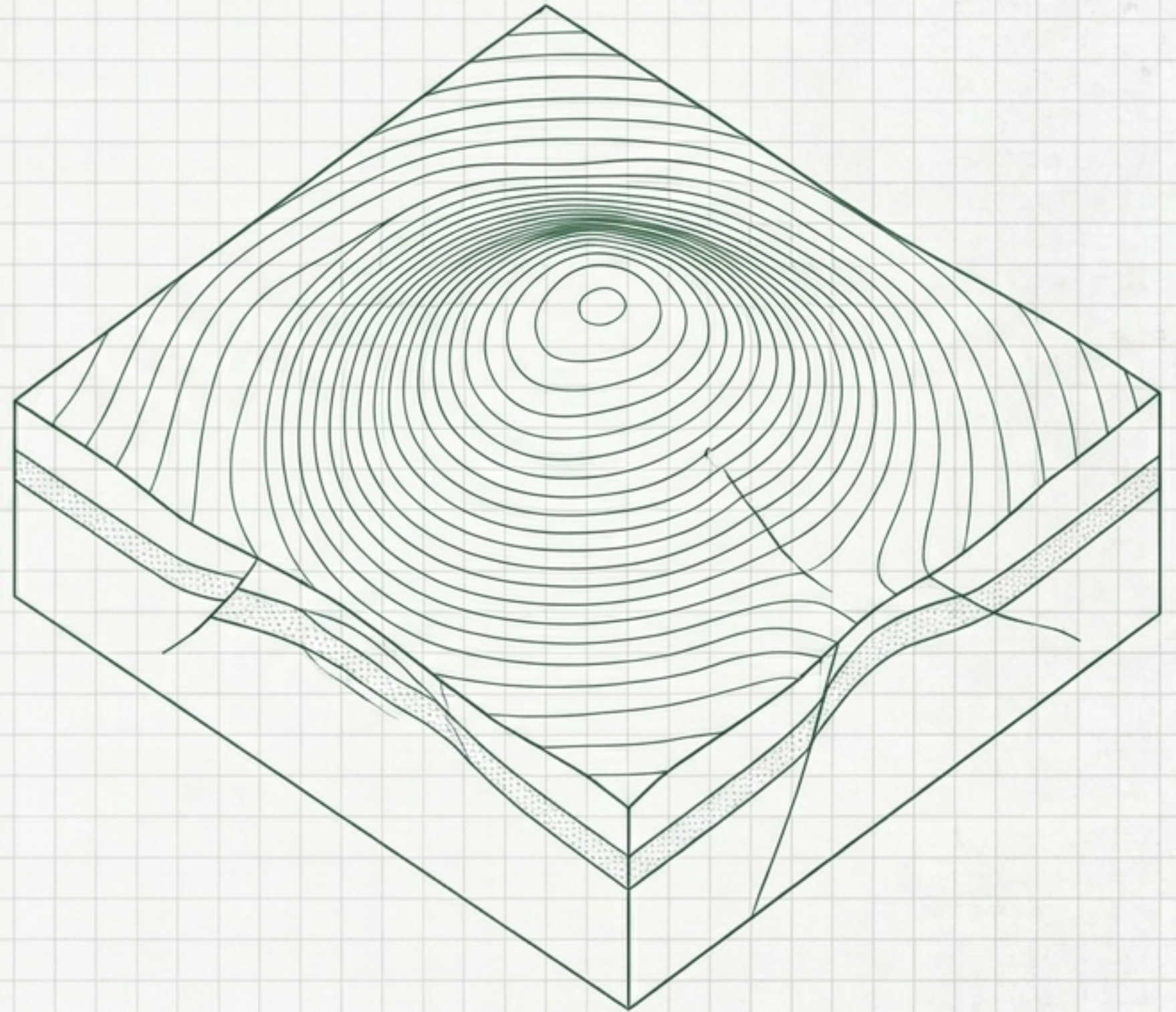
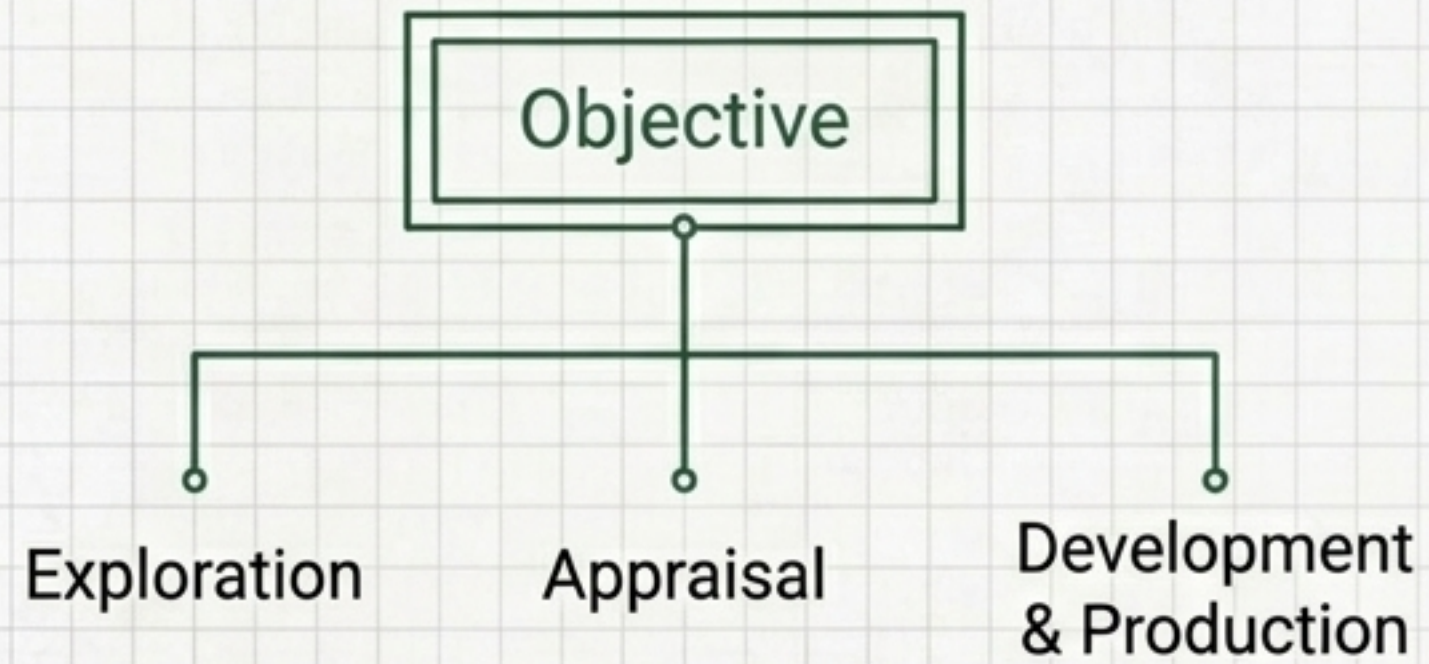


The Complexity Matrix: Optimizing Reservoir Contact

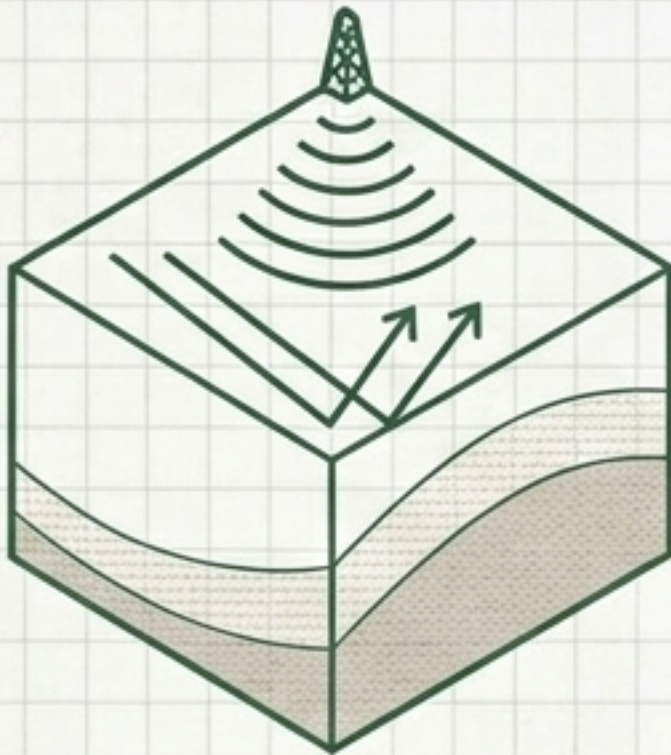


Dimension Two: Objective

The Purpose and Lifecycle
of the Wellbore.



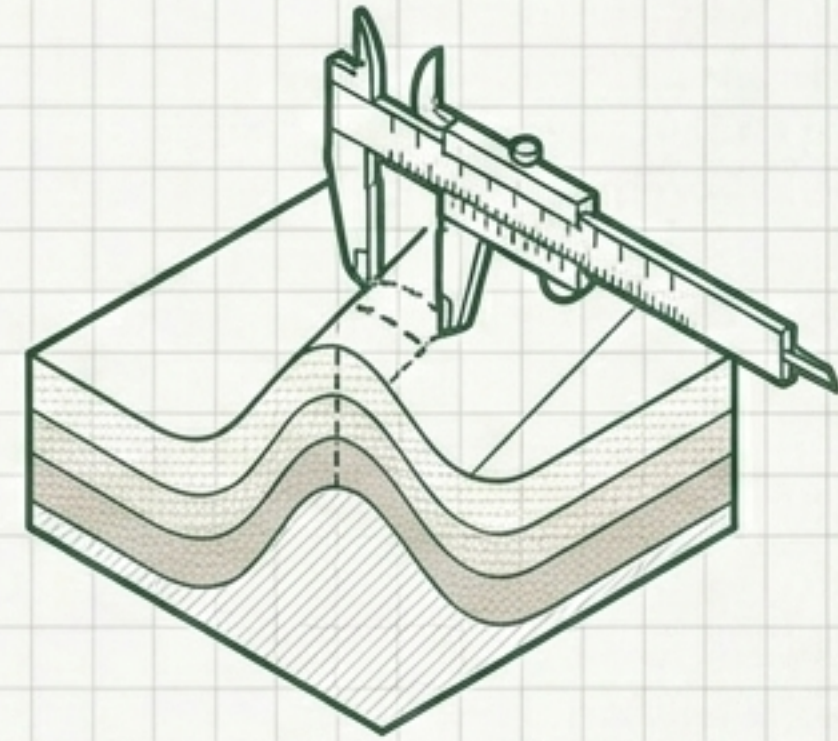
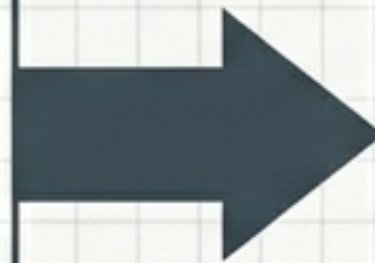
The Discovery Phase: From Wildcat to Proven Reserves



Phase 1: Exploration Well

Often called "Wildcat" wells. Drilled in unproven, frontier areas with high geological risk.

- Goal: Prove the existence of a working hydrocarbon system and validate initial seismic models.



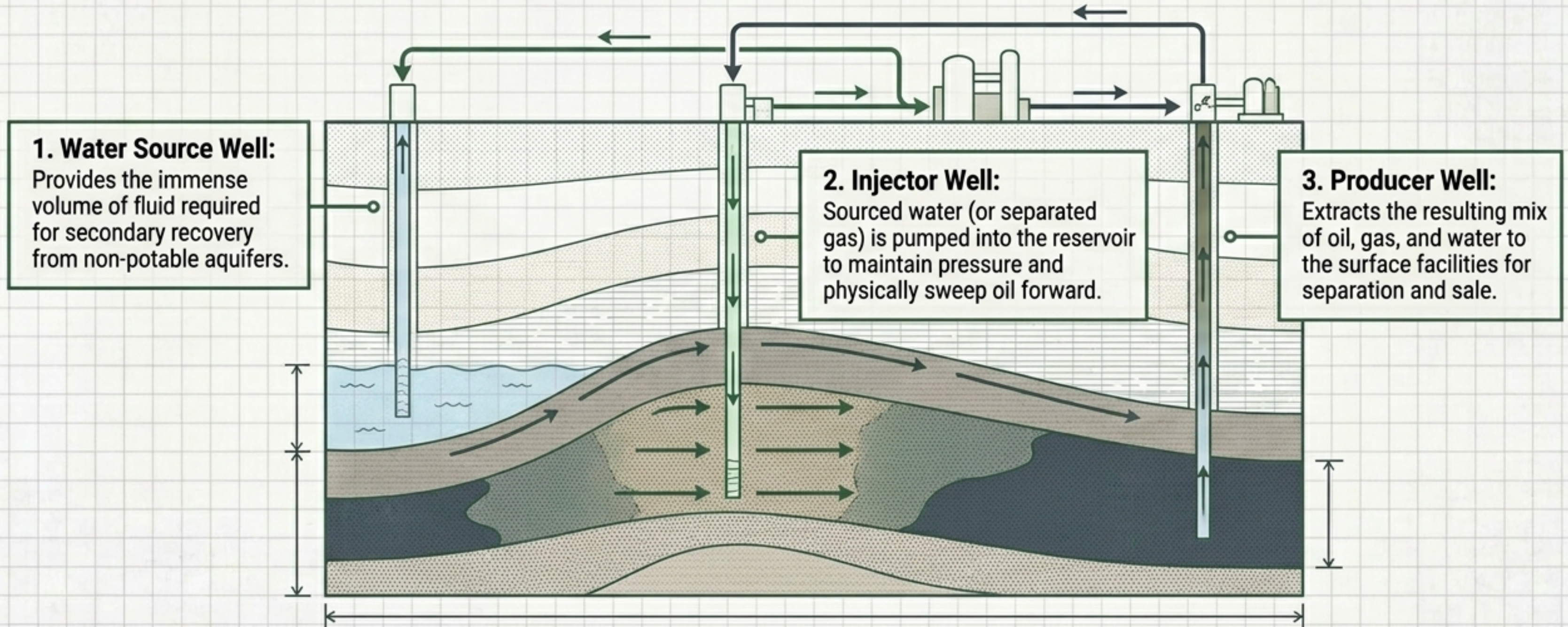
Phase 2: Appraisal Well

Drilled strategically after a successful initial exploration well.

- Goal: Delineation. Determine the physical extent, boundaries, reserve volume, and commercial viability before committing to development.

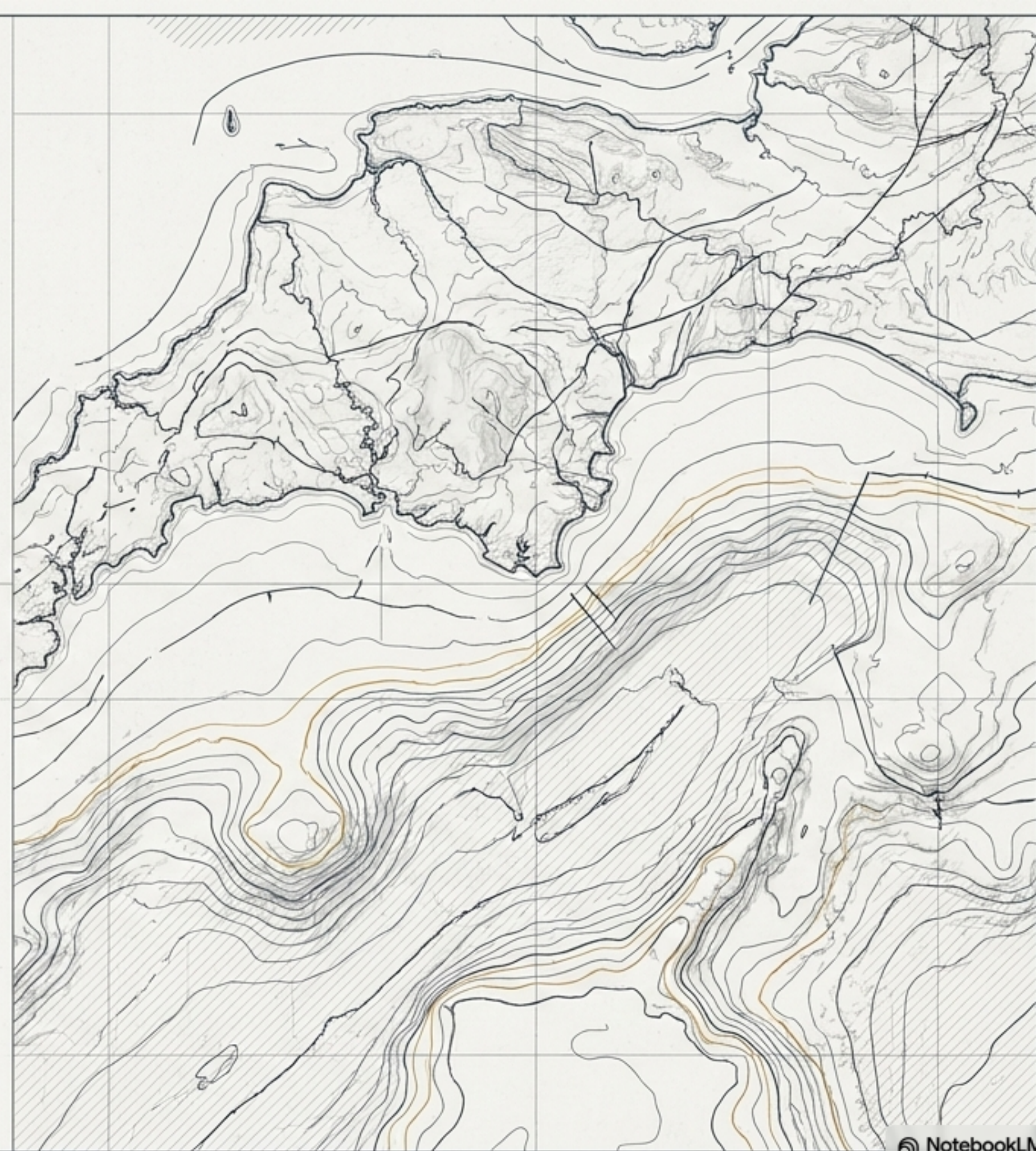
The Development Ecosystem: Extraction and Maintenance

A mature field is a balanced pressure system, not just a series of extraction straws.



Dimension Three: Environments

The Topographical and
Logistical Realities.

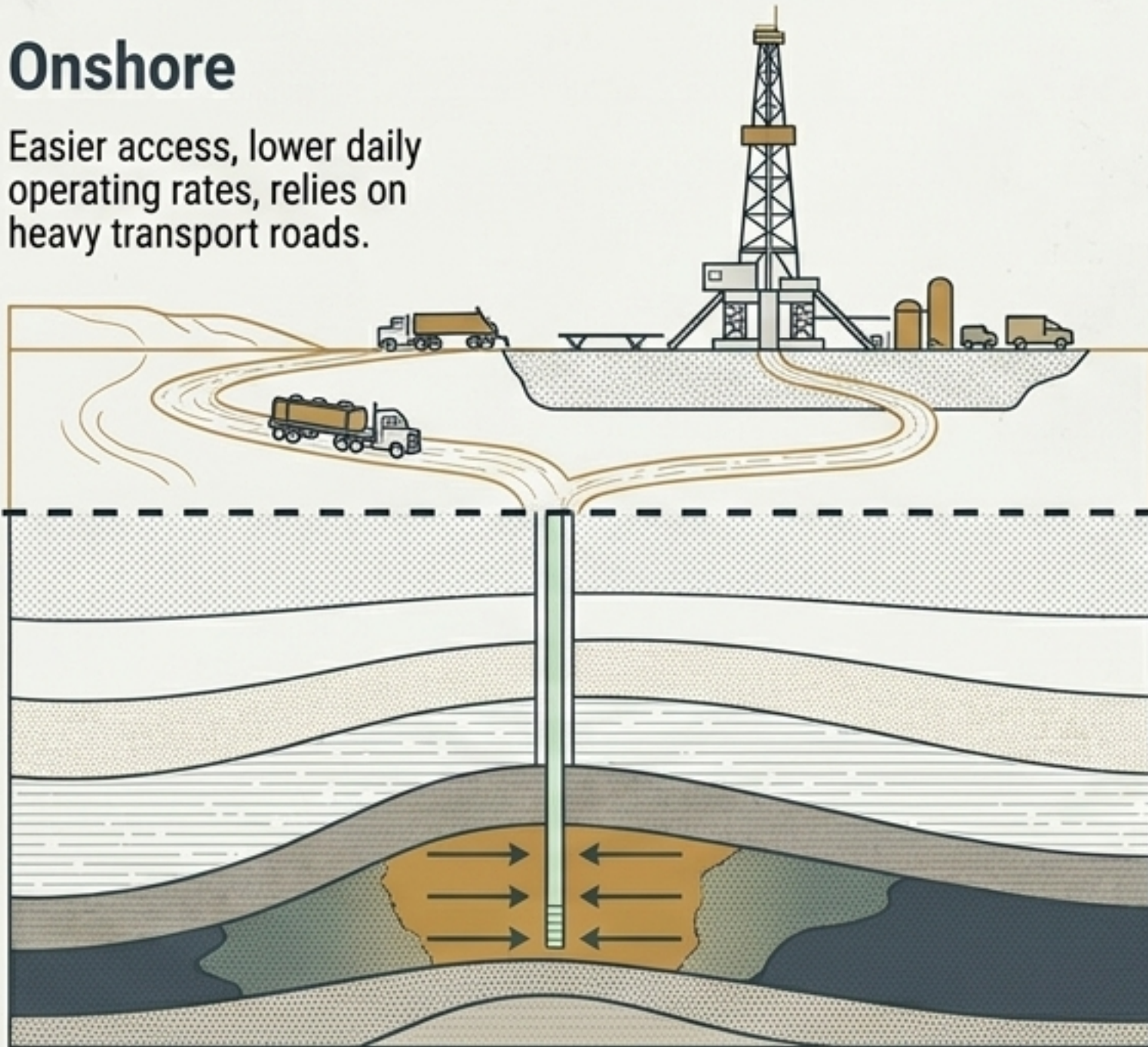


The Surface-to-Subsurface Split

The environment dictates surface logistics, but fundamental reservoir physics remain constant.

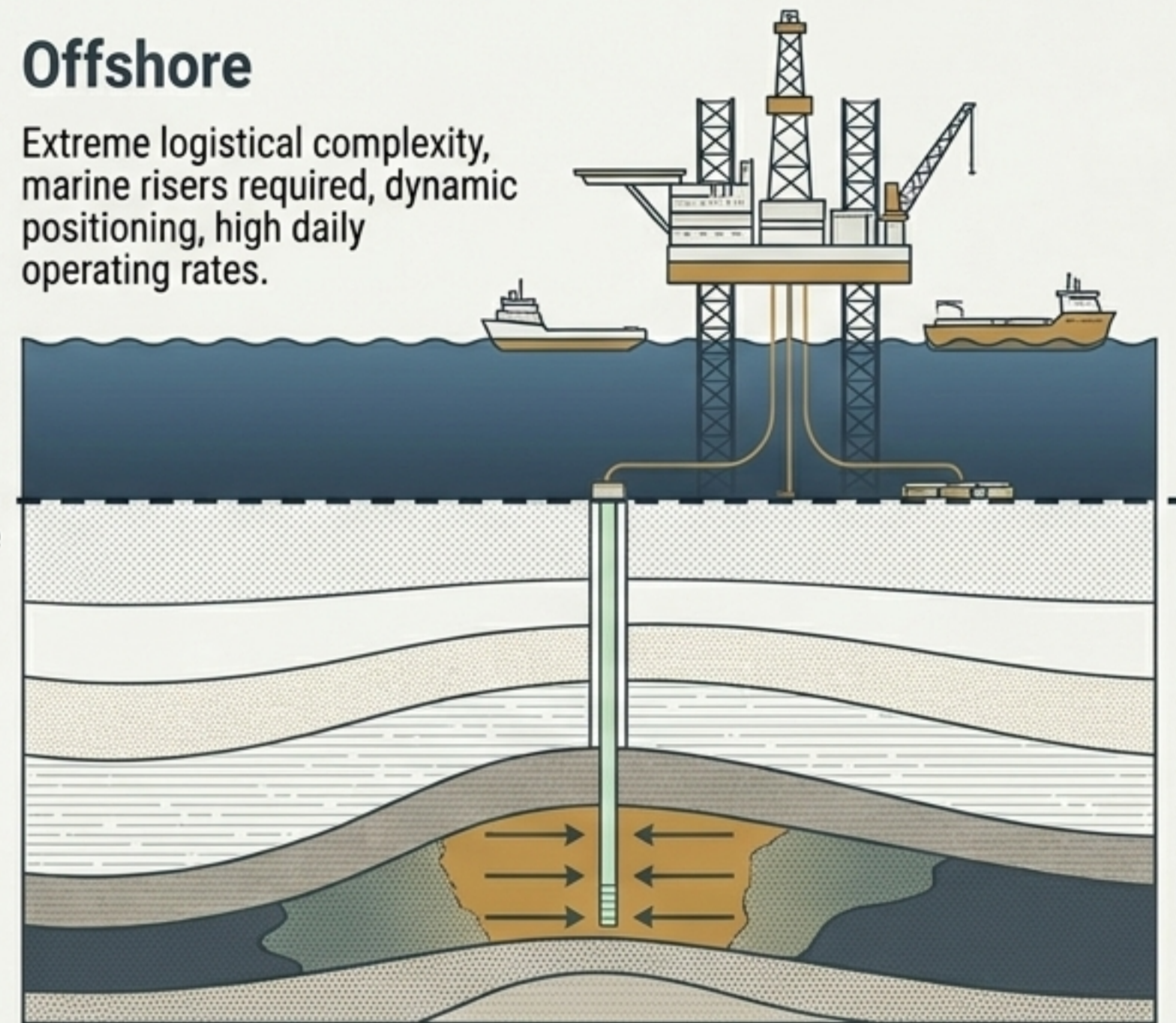
Onshore

Easier access, lower daily operating rates, relies on heavy transport roads.

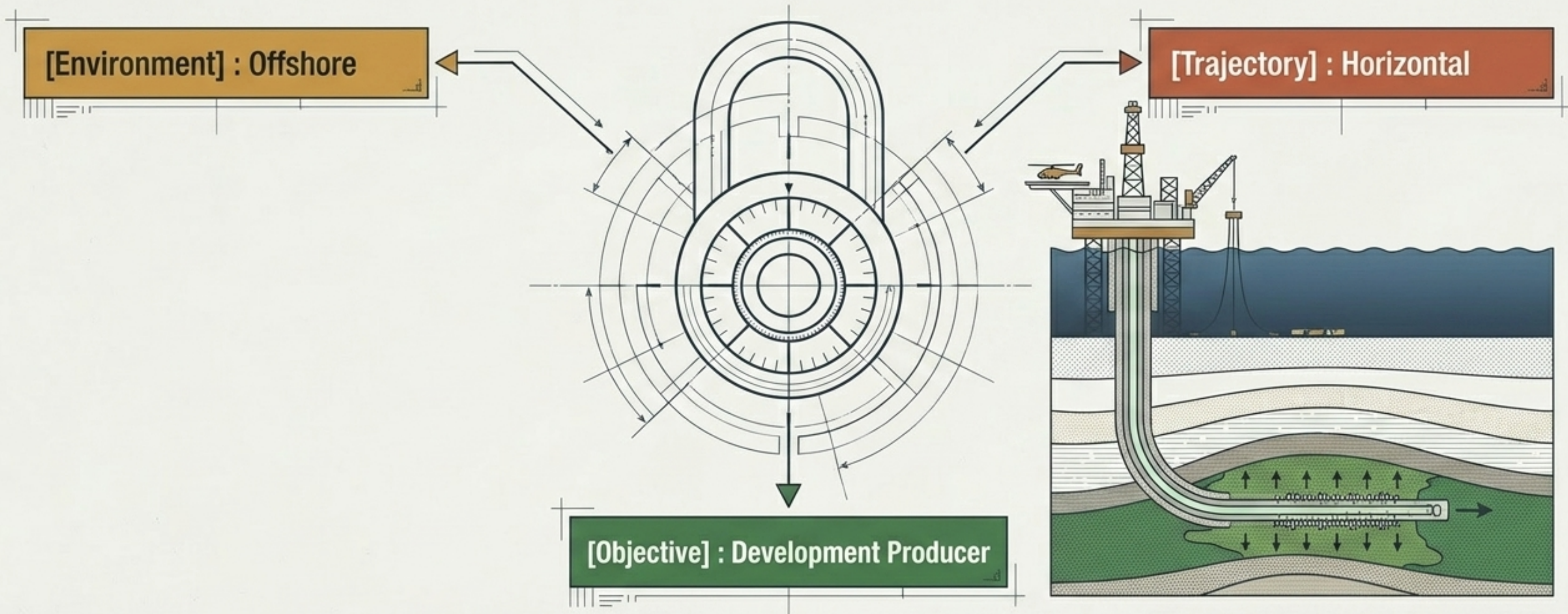


Offshore

Extreme logistical complexity, marine risers required, dynamic positioning, high daily operating rates.

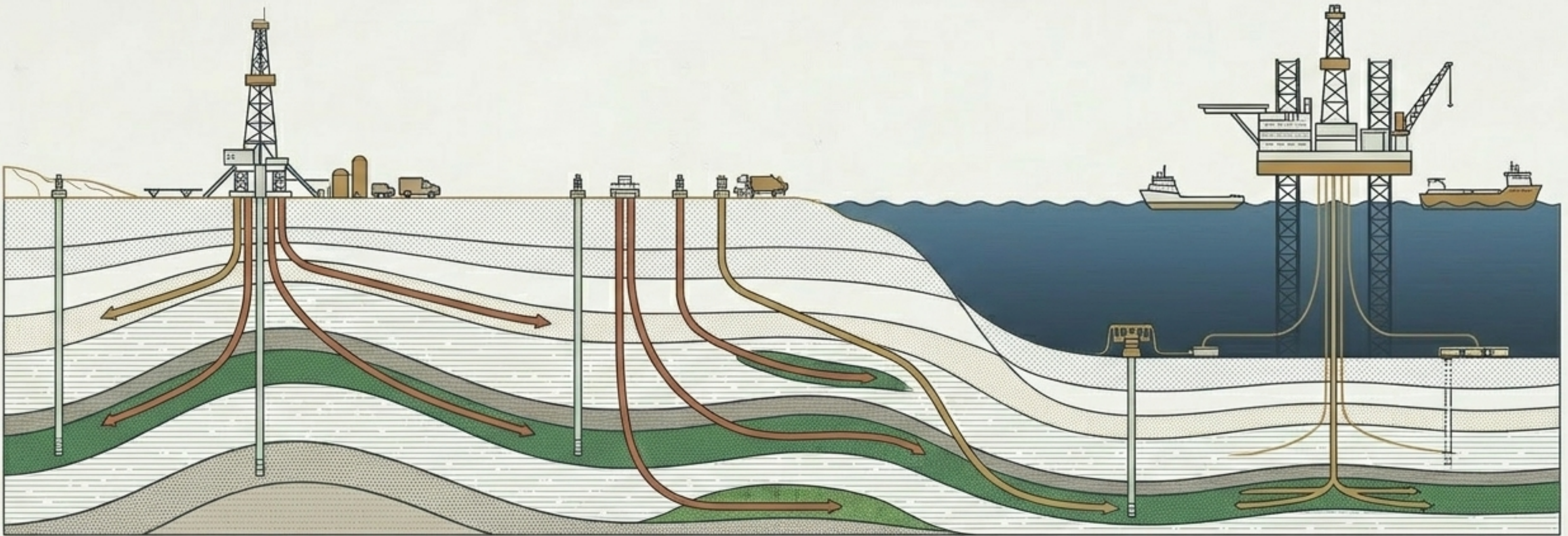


The Synthesis: Defining the Complete Well Profile



In practice, no classification exists in isolation. A modern drilling engineer must synthesize all three dimensions simultaneously to execute a successful drilling program.

Mastery of the Subsurface



Effective drilling engineering requires precise alignment of Geometry (Trajectory), Purpose (Objective), and Logistics (Environment).